

AeroTwin

Vector DB Final Knowledge Base

Source-grounded knowledge for UAV piston-engine health monitoring & predictive maintenance

Final formatted documentation

AeroTwin Vector Database — Final Knowledge Base

1. Purpose

AeroTwin is an AI-driven Digital Twin based system for UAV piston-engine health monitoring and predictive maintenance.

The system combines:

- Machine Learning models
- Physics-based monitoring
- Health Fusion
- Digital Twin
- Retrieval-Augmented Generation (RAG)
- LLM-based Maintenance Copilot

The objective is to monitor engine health, detect abnormal conditions, estimate Remaining Useful Life (RUL), analyze vibration/bearing health, generate supporting predictive-maintenance signals, combine multiple health indicators, and provide explainable maintenance-oriented information.

Knowledge type: AeroTwin project knowledge

Source: AeroTwin project documentation

2. AeroTwin System Architecture

AeroTwin follows this conceptual pipeline:

Engine Sensor Data → Data Preprocessing → ML Models → Physics Model → Health Fusion → Digital Twin → RAG → LLM Maintenance Copilot

The ML layer contains four major models:

Fault Detection Model

RUL Prediction Model

Bearing and Vibration Health Model

Auxiliary Predictive Maintenance Model

The Physics Model provides a physical-consistency or deviation signal.

Health Fusion combines multiple health signals.

The Digital Twin represents the monitored engine's current health and operational state.

The RAG and LLM layer uses technical/project knowledge together with current AeroTwin results to provide explanations.

Knowledge type: AeroTwin project knowledge

Source: AeroTwin project documentation

3. AeroTwin Model Knowledge

3.1 Model 1 — Fault Detection

Model 1 is the primary Fault Detection Model of AeroTwin.

Its purpose is to identify whether the engine is operating normally or showing a fault-related condition based on available engine sensor information.

The model receives preprocessed engine parameters and generates a fault-related prediction.

Its output is one input to Health Fusion.

Model 1 should not be treated as the only source of engine-health information because AeroTwin combines multiple health signals.

Knowledge type: model knowledge

Source: AeroTwin project documentation

3.2 Model 2 — RUL Prediction

Model 2 is the Remaining Useful Life (RUL) Prediction Model.

Its purpose is to estimate the remaining useful operating life of the monitored engine or component.

RUL is a prognostic output rather than only a current fault-state prediction.

The RUL output contributes to Health Fusion and can be used by the Maintenance Copilot to explain the current prognostic condition.

The **current RUL value must be retrieved dynamically from the deployed model/backend**. It should not be stored as a permanent static RAG fact.

Knowledge type: model knowledge

Source: AeroTwin project documentation

3.3 Model 3 — Bearing and Vibration Health

Model 3 analyzes vibration-related engine behavior and provides a supporting indication of bearing or vibration health.

Its output contributes to Health Fusion.

Current vibration measurements and current Model 3 predictions should be retrieved dynamically from the AeroTwin backend when answering questions about the current engine condition.

Knowledge type: model knowledge

Source: AeroTwin project documentation

3.4 Model 4 — Auxiliary Predictive Maintenance

Model 4 is an auxiliary predictive-maintenance model.

It is a supporting signal and should not be treated as the sole engine-fault detector.

The transfer-learning strategy uses an initial predictive-maintenance model trained on the AI4I 2020 dataset and adapts the learned representation toward the engine domain.

A neural-network architecture is used because neural networks provide a conventional fine-tuning mechanism through their learned weights.

Knowledge type: model knowledge

Source: AeroTwin project documentation

4. AI4I 2020 Dataset Knowledge

The AI4I 2020 Predictive Maintenance dataset contains 10,000 records.

The original Machine failure target is binary:

- 0 = No machine failure
- 1 = Machine failure

Class distribution:

- 9,661 normal records
- 339 failure records

For the AeroTwin Model 4 experiment, selected sensor features were:

- Air temperature [K]
- Process temperature [K]
- Rotational speed [rpm]
- Torque [Nm]
- Tool wear [min]

Direct failure-mode columns should not be blindly used as predictive inputs for Machine failure, because they are directly associated with the failure target and can introduce target leakage.

Knowledge type: dataset knowledge

Source: AeroTwin experiment documentation

5. Model 4 AI4I Experiment

Architecture:

Input → Dense(64, ReLU) → Batch Normalization → Dropout(0.30) → Dense(32, ReLU) → Dropout(0.20) → Dense(16, ReLU) → Dense(1, Sigmoid)

Training configuration:

- Optimizer: Adam
- Learning rate: 0.001
- Loss: Binary Cross Entropy
- Early stopping: Validation PR-AUC

An earlier experiment produced:

- Accuracy: 94.40%
- Precision: 37.21%
- Recall: 94.12%
- F1 Score: 53.33%
- ROC-AUC: 98.71%
- PR-AUC: 83.67%

Confusion matrix:

- True Negative: 1824
- False Positive: 108
- False Negative: 4
- True Positive: 64

Important limitation: these results came from an earlier experiment in which the test set was also used as validation data for training/early stopping. Therefore, these metrics must be treated as experimental results and not as a final clean holdout evaluation.

A rigorous evaluation should use:

- 70% training
- 15% validation
- 15% test

SMOTE should be applied only to the training data. Validation should be used for early stopping and threshold tuning. The final test set should remain untouched until final evaluation.

Knowledge type: experiment knowledge

Source: AeroTwin experiment documentation

6. Initial Engine Dataset Knowledge

The initially investigated `engine_failure_dataset.csv` contained:

- 1000 rows
- 11 columns

Features included:

- Temperature (°C)
- RPM
- Fuel_Efficiency
- Vibration_X
- Vibration_Y
- Vibration_Z
- Torque
- Power_Output (kW)
- Operational_Mode

Target:

- Fault_Condition

Fault_Condition contained four classes:

- Class 0: 258
- Class 1: 248

- Class 2: 265
- Class 3: 229

Initial baseline results:

- DummyClassifier: approximately 26.5% accuracy
- Logistic Regression: approximately 27.5%
- Random Forest: approximately 29.5%
- Engineered Random Forest: approximately 27.0%

The predictive signal was weak. Therefore, the dataset's label semantics, provenance, and suitability for transfer learning were questioned.

This dataset should not automatically be treated as the final validated engine-domain dataset.

Knowledge type: dataset limitation / experiment knowledge

Source: AeroTwin experiment documentation

7. Health Fusion

Health Fusion provides an integrated engine-health assessment by combining multiple AeroTwin health signals.

Potential inputs include:

- Fault Detection prediction
- RUL prediction
- Bearing/Vibration health
- Auxiliary Predictive Maintenance signal
- Physics-model deviation

Conceptually:

ML Health Signals + Physics-Based Information → Health Fusion → Integrated Engine Health State

Health Fusion should preserve contributing signals so that the Maintenance Copilot can explain why an overall health state was produced.

Health Fusion is a structured health-assessment layer. It is not the LLM and does not replace the individual ML models.

Knowledge type: AeroTwin architecture

Source: AeroTwin project documentation

8. Physics-Based Monitoring in AeroTwin

Physics-based monitoring evaluates system behavior using physical relationships, mathematical models, engineering relationships, or expected system behavior.

A simplified concept is:

Expected System Behavior – Observed System Behavior = Residual / Deviation

A deviation can provide a supporting indication of abnormal system behavior.

In AeroTwin, the Physics Model produces a physical-consistency or deviation signal that contributes to Health Fusion.

This is an AeroTwin design concept. The provided FAA/NASA source documents do not define a specific UAV piston-engine physics equation or physics-model algorithm.

Knowledge type: AeroTwin design knowledge

Source: AeroTwin project documentation; source limitation noted

9. Digital Twin

The AeroTwin Digital Twin represents the monitored engine's current health and operational state.

It receives information from:

- Sensor processing
- ML models
- Physics-based analysis
- Health Fusion

The Digital Twin can represent:

- Current operating condition
- Detected abnormalities
- Predicted health
- RUL
- Vibration condition
- Physics deviation
- Overall health state

The current Digital Twin state must be retrieved dynamically from the backend when answering questions about the current engine.

Knowledge type: AeroTwin architecture

Source: AeroTwin project documentation

10. RAG and Maintenance Copilot

The AeroTwin RAG layer provides technical and maintenance knowledge to the Maintenance Copilot.

RAG retrieves relevant information from the knowledge base and supplies it as context to the LLM before response generation.

The purpose of RAG is to provide grounded technical and maintenance information rather than relying only on the LLM's internal knowledge.

RAG does not replace:

- ML models
- Physics Model
- Health Fusion
- Digital Twin

The LLM uses retrieved knowledge and current AeroTwin results to generate a natural-language explanation.

Knowledge type: AeroTwin architecture

Source: AeroTwin project documentation

11. Static Knowledge vs Current Engine Data

Static information — Vector Database

The Vector Database should contain relatively stable information such as:

- Technical definitions
- Engineering concepts
- Maintenance concepts
- FAA IAHM concepts
- NASA RCM concepts
- AeroTwin architecture
- Model descriptions
- Dataset descriptions

- Experiment results
- Experiment limitations
- Failure-mode concepts
- Maintenance strategies
- Source-grounded technical documentation

Dynamic information — Backend / PostgreSQL / Model APIs

Current engine information should be retrieved dynamically:

- Current temperature
- Current RPM
- Current vibration
- Current fuel efficiency
- Current fault prediction
- Current RUL
- Current bearing health
- Current physics deviation
- Current Health Fusion result
- Current Digital Twin state

Therefore:

Static Technical / Project Knowledge → Vector Database

Current Engine State → Backend / PostgreSQL / Model APIs

The LLM can combine both sources when answering a question.

Knowledge type: system/data-governance knowledge

Source: AeroTwin architecture

12. Engine Monitoring

Engine monitoring is the continuous or periodic observation of engine operating parameters to understand the current operating and health condition of an engine.

Possible monitoring parameters include:

- Temperature
- RPM
- Pressure

- Torque
- Fuel efficiency
- Fuel flow
- Vibration
- Power output
- Oil temperature
- Oil pressure
- Operational mode

Engine monitoring can use threshold-based, trend-based, statistical, machine-learning, or physics-based approaches.

Important: these parameter categories are general monitoring concepts. They are not claimed to be mandatory or complete sensor requirements for a UAV piston engine.

Knowledge type: technical knowledge

Source: AeroTwin project knowledge; source-specific engine parameter limits are not provided

13. Fault Detection

Fault detection is the process of determining whether an abnormal or faulty condition exists in a system.

Fault detection answers:

"Is the system showing an abnormal condition?"

Possible approaches include:

- Threshold-based methods
- Statistical methods
- Machine learning
- Deep learning
- Physics-based residual analysis
- Hybrid ML and physics approaches

Fault detection should be distinguished from:

- **Fault isolation:** determining which component or subsystem is associated with a fault.
- **Fault diagnosis:** characterizing or identifying the cause and nature of a fault.

In AeroTwin, Model 1 provides the primary fault-detection signal.

Knowledge type: technical + AeroTwin project knowledge

Source: AeroTwin project documentation

14. Prognostics and RUL

Prognostics is the process of estimating future health or degradation.

Remaining Useful Life (RUL) is an estimate of remaining useful operating life before a defined failure or degradation condition is reached.

Fault detection focuses mainly on the current condition.

Prognostics focuses on future condition and remaining life.

In AeroTwin, Model 2 provides RUL-related prognostic information.

The provided NASA RCM Guide discusses the conceptual progression:

Detect → Alert → Confident RUL → Remove

It also discusses prediction error bounds.

The source does **not** provide a specific RUL equation or algorithm. Therefore, no NASA RUL formula should be claimed for AeroTwin.

Knowledge type: technical + source-grounded knowledge

Source: NASA RCM Guide, §4.1, Figure 4-3, p. 4-3; AeroTwin project documentation

15. Predictive Maintenance

Predictive maintenance uses equipment condition, historical information, sensor measurements, and predictive models to anticipate degradation or potential failure and support maintenance planning.

A simplified flow is:

Sensor Data → Health Assessment → Prediction → Maintenance Planning

Possible technologies include:

- Machine learning
- Deep learning
- Time-series analysis
- Vibration analysis

- Statistical methods
- Physics-based models
- Digital twins

AeroTwin uses multiple health signals to support predictive-maintenance analysis.

Knowledge type: technical knowledge

Source: AeroTwin project documentation; NASA RCM Guide for PT&I concepts

16. Condition-Based Maintenance

Condition-Based Maintenance (CBM) makes maintenance decisions based on observed equipment condition.

A simplified process is:

Condition Monitoring → Health Assessment → Maintenance Decision

Condition information can include:

- Increasing vibration
- Increasing temperature
- Abnormal pressure
- Reduced performance
- Increasing degradation
- Fault detection
- Reduced predicted remaining life

CBM differs from fixed scheduled maintenance because the maintenance decision is linked to observed equipment condition.

NASA describes Predictive Testing & Inspection (PT&I) as a condition-monitoring approach using non-intrusive testing, visual inspection, and performance data to assess machinery condition and plan maintenance before failure.

Knowledge type: maintenance knowledge

Source: NASA RCM Guide, §5.3, p. 5-7

17. Reliability-Centered Maintenance (RCM)

Reliability-Centered Maintenance is a structured approach for determining maintenance strategies based on:

- System functions
- Functional failures
- Failure modes
- Failure effects
- Consequences
- Available maintenance tasks

A simplified reasoning flow is:

System Function → Functional Failure → Failure Mode → Failure Effect → Consequence → Maintenance Strategy

RCM asks questions such as:

- What function must the system perform?
- How can that function fail?
- What can cause the failure?
- What happens when the failure occurs?
- Can the failure be detected?
- What maintenance action can manage the failure?

NASA states that RCM integrates preventive maintenance, predictive testing and inspection, repair/reactive maintenance, and proactive maintenance.

Knowledge type: maintenance knowledge

Source: NASA RCM Guide, §§3.1, 3.3.1.2

18. Integrated Aircraft Health Management (IAHM)

FAA Advisory Circular AC 43-218 describes Integrated Aircraft Health Management (IAHM) as an aircraft health-monitoring approach using onboard sensors, data transmission, and data analysis to provide information about aircraft system performance and structural condition.

The FAA document describes four end-to-end architectural components:

Aircraft Systems

Data Transmission

Data Analysis and Implementation

Ground-Based Equipment

The document also discusses:

- System criticality

- Missing or latent data
- Alert levels
- Fault-detection performance
- Trained personnel
- Data retention and security
- Maintenance-program data use

Important scope limitation: AC 43-218 concerns FAA authorization of IAHM for specified certificated-aircraft operators. It should not be interpreted as a UAV-specific regulatory requirement for AeroTwin.

Knowledge type: source-grounded technical knowledge

Source: FAA AC 43-218, §§1, 5, 8, 9, 10

19. FAA Alert-Level Concept

FAA AC 43-218 defines an alert level as a predetermined value representing system degradation established during system certification.

The document explains that operators may adjust alert levels to identify trends or wear before a valid fault.

This provides a useful general health-monitoring concept:

Degradation Trend → Alert → Maintenance Assessment

However, AeroTwin must not claim that an FAA alert threshold directly defines its ML threshold.

Any AeroTwin-specific threshold must be derived and validated using AeroTwin data and engineering evidence.

Knowledge type: source-grounded knowledge + AeroTwin design distinction

Source: FAA AC 43-218, ¶9.7.2

20. Fault Detection Performance

FAA AC 43-218 discusses reporting cases where an IAHM system:

- Fails to detect a fault that it was intended to detect and maintenance later confirms it.
- Detects a fault that maintenance/shop testing does not confirm.

This is conceptually relevant to evaluating fault-detection systems.

For AeroTwin, model performance should be evaluated using appropriate ground-truth data and metrics such as precision, recall, F1, false positives, and false negatives.

The FAA document does not prescribe AeroTwin's ML metrics or model architecture.

Knowledge type: source-grounded knowledge + AeroTwin experiment knowledge

Source: FAA AC 43-218, ¶10.3.4; AeroTwin experiment documentation

21. Failure Modes and FMEA

Failure Mode and Effects Analysis (FMEA) examines:

- System functions
- Possible functional failures
- Failure modes
- Failure consequences
- Effects on operation or mission

NASA's RCM Guide describes FMEA at system, subsystem, and component levels.

A failure mode describes a specific way in which a component, subsystem, or system can fail to perform its intended function.

A failure effect describes what happens because of the failure.

These concepts are useful for organizing maintenance knowledge and explaining detected conditions.

Knowledge type: maintenance knowledge

Source: NASA RCM Guide, §4.2, p. 4-9

22. Failure Effects and Consequences

Failure effects describe consequences or observable results associated with a failure mode.

Consequences can be considered at different levels:

Component → Subsystem → Engine → Aircraft → Mission

Examples may include:

- Reduced engine performance
- Increased temperature
- Increased vibration

- Reduced efficiency
- Loss of component functionality
- Engine shutdown
- Increased operational or mission risk

These are conceptual failure-effect examples, not UAV-specific fault codes or validated failure mappings.

Knowledge type: maintenance knowledge

Source: NASA RCM concepts + AeroTwin knowledge

23. Vibration Monitoring

NASA's RCM Guide describes vibration analysis as a Predictive Testing & Inspection technique.

Vibration information can be analyzed for:

- Trends
- Changes from baseline
- Patterns
- Indications of degradation
- Correlation with other condition-monitoring data

NASA also describes correlation of multiple PT&I technologies to increase detection confidence.

AeroTwin Model 3 provides a supporting bearing/vibration health signal.

Important: generic vibration values in the NASA guide are not UAV piston-engine limits.

The NASA guide's example values such as 0.1 ips and 0.2 ips are illustrative figures for explaining performance margin and must not be reused as AeroTwin engine thresholds.

Knowledge type: maintenance knowledge + AeroTwin model knowledge

Source: NASA RCM Guide, §§4.1, 5.3, 6.3

24. Lubricant and Wear-Particle Analysis

NASA's RCM Guide describes lubricant analysis as a method for determining:

- Mechanical wear condition
- Lubricant condition
- Contamination

The guide also discusses correlating lubricant/wear information with vibration analysis.

The NASA guide includes a diesel-engine oil-analysis subsection. This is the closest reciprocating-engine example in the provided source set.

Important scope limitation: this is generic facilities/diesel-generator guidance, not aviation/UAV-specific guidance. Any numeric oil-viscosity example or maintenance interval from that source must not be treated as a validated UAV piston-engine limit.

Knowledge type: maintenance knowledge

Source: NASA RCM Guide, §6.6

25. Physics and Performance Degradation

NASA's System Performance Curve describes expected performance, degradation, minimum required performance, and functional failure.

Conceptually:

Expected / Required Performance → Degradation → Functional Failure

AeroTwin uses a related design idea in its Physics Model:

Expected Behavior → Compare with Observed Behavior → Deviation Signal

This is an AeroTwin design inference from the general degradation concept. The NASA guide does not define a UAV piston-engine physics model.

Knowledge type: source-grounded concept + AeroTwin design inference

Source: NASA RCM Guide, §4.1, Figure 4-1

26. P-F Curve

The P-F concept describes the interval between the point where degradation becomes detectable and the point of functional failure.

Conceptually:

Potential Failure → Detectable Degradation → Functional Failure

The P-F concept helps explain why condition monitoring should detect degradation before functional failure.

AeroTwin may use the concept when explaining degradation and prognostics, but the source does not provide an AeroTwin-specific P-F threshold or RUL algorithm.

Knowledge type: maintenance knowledge

Source: NASA RCM Guide, §4.1, Figure 4-2

27. Maintenance Strategies

Common maintenance strategies include:

Corrective / Reactive Maintenance

Maintenance occurs after a failure or unacceptable condition.

Failure → Repair / Replacement

Preventive / Scheduled Maintenance

Maintenance occurs according to predefined time, usage, or operational intervals.

Time / Usage → Scheduled Maintenance

Condition-Based Maintenance

Maintenance is based on observed equipment condition.

Condition Monitoring → Health Assessment → Maintenance

Predictive Maintenance

Maintenance planning uses predicted future condition, degradation, or failure information.

Monitoring → Prediction → Future Health / Risk → Maintenance Planning

Proactive Maintenance

Maintenance focuses on identifying and addressing underlying causes of recurring failures or degradation.

Examples include:

- Root-cause analysis
- Design improvement
- Process improvement
- Operating-condition changes

Knowledge type: maintenance knowledge

Source: NASA RCM Guide, §§3–5

28. Maintenance Decision Support

A maintenance decision can consider:

- Current health state
- Fault information
- RUL
- Vibration condition
- Sensor trends
- Physics deviation
- Failure consequences
- Operational context
- Applicable technical documentation

Possible actions can include:

- Continue monitoring
- Perform additional inspection
- Schedule maintenance
- Perform further diagnosis
- Repair a component
- Replace a component
- Take immediate operational action when required by applicable procedures

The AeroTwin Maintenance Copilot is intended to support and explain maintenance information.

It should not invent maintenance limits, procedures, inspection intervals, or engineering requirements.

Knowledge type: maintenance knowledge + AeroTwin safety rule

Source: NASA RCM Guide; AeroTwin design requirements

29. Current Engine State Query Flow

For a question such as:

"Why is the engine health abnormal?"

The system should follow:

****User Question**

- Identify Current-Health Query
- Retrieve Current AeroTwin State
- Retrieve Relevant Technical Knowledge
- Combine Current Results + Retrieved Knowledge
- LLM Explanation
- Maintenance-Oriented Response******

Current AeroTwin state may contain:

- Fault Detection result
- RUL
- Vibration/Bearing health
- Auxiliary predictive-maintenance result
- Physics deviation
- Health Fusion state
- Digital Twin state

The RAG system provides technical context. The backend provides current values.

Knowledge type: AeroTwin system architecture

30. Maintenance Copilot Responsibilities

The Maintenance Copilot can:

- Explain the current engine health state
- Explain ML model outputs
- Explain Health Fusion
- Explain detected abnormal conditions
- Explain RUL information
- Explain vibration-related health information
- Retrieve relevant maintenance concepts
- Explain failure modes and failure effects
- Explain predictive-maintenance concepts
- Answer questions about AeroTwin architecture
- Summarize available engine-health information

- Connect current observations with retrieved technical knowledge

The Maintenance Copilot must not:

- Invent sensor values
- Invent RUL values
- Invent fault probabilities
- Invent health scores
- Invent engineering limits
- Invent maintenance intervals
- Invent maintenance procedures
- Override model predictions without evidence
- Present unsupported information as established fact
- Replace approved maintenance documentation or procedures

31. Source Priority and Grounding

When multiple sources provide information:

Applicable authoritative technical/maintenance documentation should take priority for engineering procedures, limitations, and maintenance requirements.

AeroTwin project knowledge should be used for AeroTwin architecture, models, datasets, and experiments.

Live AeroTwin backend should be used for current engine values and current model outputs.

The LLM should clearly distinguish retrieved facts from model-generated interpretation.

Generated responses are decision-support explanations and should not replace applicable approved technical or maintenance documentation.

32. Source Limitations

The current FAA/NASA source set does **not** establish:

- UAV-specific engine construction details
- UAV piston-engine operating limits
- RPM limits
- Temperature limits
- Pressure limits
- UAV-specific vibration thresholds

- UAV-specific fault codes
- UAV-specific maintenance intervals
- UAV sensor/telemetry specifications
- AeroTwin-specific RUL equations
- AeroTwin-specific RUL algorithms
- Aircraft/UAV-specific bearing-life limits
- Approved UAV maintenance procedures

The FAA AC 43-218 document concerns IAHM authorization for specified certificated-aircraft operators and should not be presented as a UAV regulatory requirement.

The NASA RCM Guide is primarily for NASA facilities and collateral equipment. Its diesel/reciprocating-engine material is a conceptual analogue and should not be treated as aircraft/UAV-specific.

Knowledge type: source limitation

Sources: FAA AC 43-218; NASA RCM Guide; AeroTwin source review

33. Important Numerical-Value Rule

The following source values must **not** be used as AeroTwin target thresholds without independent validation:

- NASA's illustrative vibration values such as 0.1 ips and 0.2 ips
- NASA's generic 2-sigma / 3-sigma alert/alarm example
- NASA's reported vibration Pd/Pfa statistics
- Generic diesel-engine oil-viscosity change examples
- Any generic bearing-life numerical values

These values are source-specific examples or statistics, not validated AeroTwin UAV piston-engine limits.

Knowledge type: safety / grounding rule

34. AeroTwin Knowledge Categories

The Vector Database can organize knowledge using these categories:

Technical Knowledge

- Engine Monitoring
- Fault Detection
- Fault Isolation

- Fault Diagnosis
- Prognostics
- RUL
- Predictive Maintenance
- Condition-Based Maintenance
- Reliability-Centered Maintenance
- Integrated Aircraft Health Management
- Physics-Based Monitoring
- Health Assessment
- Digital Twin

Maintenance Knowledge

- Failure Modes
- Failure Effects
- Failure Consequences
- Maintenance Decisions
- Corrective Maintenance
- Preventive Maintenance
- Condition-Based Maintenance
- Predictive Maintenance
- Proactive Maintenance
- Maintenance Planning
- Vibration Monitoring
- Lubricant / Wear Analysis
- FMEA
- P-F Curve

AeroTwin Project Knowledge

- Project Objective
- System Architecture
- Model 1
- Model 2
- Model 3
- Model 4
- Health Fusion
- Physics Model
- Digital Twin
- RAG
- Maintenance Copilot

Dataset Knowledge

- AI4I 2020
- Engine Failure Dataset
- NGAFID Candidate Dataset
- Feature Descriptions
- Target Definitions
- Class Distribution
- Data Limitations
- Dataset Suitability

Experiment Knowledge

- Model Architecture
- Training Strategy
- Validation Strategy
- Test Strategy
- Evaluation Metrics
- Confusion Matrix
- Transfer Learning
- SMOTE
- Target Leakage
- Experiment Limitations

35. NGAFID Dataset Status

NGAFID is a candidate aviation-domain dataset in the AeroTwin project.

It should be treated as a candidate source until its metadata, time-series relationship, labels, and intended target are fully verified.

The Vector Database should not present NGAFID as the final validated AeroTwin engine dataset unless the project documentation confirms that status.

Knowledge type: dataset status / limitation

Source: AeroTwin project documentation

36. RAG Answering Principle

For each Maintenance Copilot response, distinguish:

1. Technical Knowledge

Retrieved from the Vector Database.

2. AeroTwin Project Knowledge

Retrieved from the project knowledge base.

3. Current Engine Data

Retrieved from the live AeroTwin backend.

4. Model Interpretation

Generated by combining the first three sources.

The system should clearly distinguish retrieved facts from generated interpretation.

37. Final Architecture

User



Maintenance Copilot



RAG Knowledge Base

- FAA IAHM concepts
- NASA RCM knowledge
- Technical knowledge
- Maintenance knowledge
- AeroTwin project knowledge
- Dataset knowledge
- Experiment knowledge

PLUS

Live AeroTwin Backend

- Sensor data
- Model predictions
- RUL
- Vibration health

- Physics deviation
- Health Fusion
- Digital Twin state

↓

LLM

↓

Grounded Maintenance Explanation

38. Source Reference Index

FAA

AC_43-218_FAA_Web . pdf

FAA Advisory Circular 43-218, *Operational Authorization of Integrated Aircraft Health Management Systems*, 7/8/2022.

Useful topics:

- IAHM definition
- Health-monitoring architecture
- Data transmission
- Data analysis
- Ground-based equipment
- Alert levels
- Missing data
- Fault-detection performance
- Data retention/security
- Maintenance-program integration

NASA

nasa_rcmguide . pdf

NASA, *Reliability-Centered Maintenance Guide for Facilities and Collateral Equipment*, September 2008.

Useful topics:

- RCM
- FMEA
- Failure modes
- Failure effects

- Criticality
- Probability of occurrence
- P-F curve
- RUL concept
- PT&I
- Predictive maintenance
- Condition monitoring
- Vibration monitoring
- Lubricant/wear analysis
- Maintenance strategies
- Maintenance decision process

FAA Powerplant Handbook

01_amtp_front.pdf

FAA-H-8083-32B, *Aviation Maintenance Technician Handbook–Powerplant*, 2023.

Scope note: only front matter was available in the supplied source. It describes topics covered by the full handbook but does not provide the actual engine technical content. Therefore it should not be used as evidence for detailed engine-component knowledge.

39. Final Grounding Rule

AeroTwin's RAG system must never convert a general FAA/NASA concept into a UAV-specific engineering requirement unless the project has an appropriate validated source.

The system should use:

FAA/NASA → General source-grounded concepts

AeroTwin documentation → Project-specific knowledge

Backend / Model APIs → Current engine state

LLM → Explanation and reasoning over retrieved evidence

This separation is required to keep the Maintenance Copilot grounded and reduce unsupported technical claims.